# C17 Thermodynamics – Knowledge Organiser: Self-Test

*AQA A-level Chemistry • Set 1: Born–Haber Definitions* **Paper 1**

| # | Question | Attempt 1 | Attempt 2 | Attempt 3 | Answer |
|---|---|---|---|---|---|
| 1 | Enthalpy of atomisation | | | | Enthalpy change when 1 mol of gaseous atoms forms from the element in its standard state |
| 2 | First ionisation energy | | | | Enthalpy change to remove 1 mol of electrons from 1 mol of gaseous atoms |
| 3 | First electron affinity | | | | Enthalpy change when 1 mol of gaseous atoms each gain 1 electron to form 1 mol of gaseous 1− ions |
| 4 | Second electron affinity | | | | Enthalpy change when 1 mol of gaseous 1− ions each gain 1 electron to form 1 mol of gaseous 2− ions |
| 5 | Sign of first electron affinity | | | | Negative |
| 6 | Sign of second electron affinity | | | | Positive |
| 7 | Lattice enthalpy of formation | | | | Enthalpy change when 1 mol of ionic solid forms from its gaseous ions |
| 8 | Lattice enthalpy of dissociation | | | | Enthalpy change when 1 mol of ionic solid separates into its gaseous ions |
| 9 | Enthalpy of hydration | | | | Enthalpy change when 1 mol of gaseous ions dissolves in water |
| 10 | Enthalpy of solution | | | | Enthalpy change when 1 mol of solute dissolves completely in water |

*Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.*

# C17 Thermodynamics – Knowledge Organiser: Self-Test

*AQA A-level Chemistry • Set 2: Lattice Enthalpy and Entropy* Paper 1

| # | Question | Attempt 1 | Attempt 2 | Attempt 3 | Answer |
|---|---|---|---|---|---|
| 1 | Bond dissociation enthalpy | | | | Enthalpy change to break 1 mol of a covalent bond, all species gaseous |
| 2 | Equation for enthalpy of solution | | | | $\Delta_{sol}H = \Delta_{L,diss}H + \Sigma\Delta_{hyd}H$ |
| 3 | Sign of enthalpy of hydration | | | | Negative |
| 4 | Two assumptions of the perfect ionic model | | | | Ions are perfect spheres; bonding is purely ionic |
| 5 | Experimental lattice enthalpy larger than theoretical means | | | | Covalent character |
| 6 | Two factors giving a larger lattice enthalpy | | | | Smaller ions, higher charges |
| 7 | Entropy ($S$) | | | | Measure of disorder |
| 8 | Units of entropy | | | | $J\,K^{-1}\,mol^{-1}$ |
| 9 | Order of entropy for states | | | | Gas > liquid > solid |
| 10 | Entropy of a perfect crystal at 0 K | | | | 0 |

*Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.*

# C17 Thermodynamics – Knowledge Organiser: Self-Test

*AQA A-level Chemistry • Set 3: Entropy and Gibbs Free Energy* **Paper 1**

| # | Question | Attempt 1 | Attempt 2 | Attempt 3 | Answer |
|---|---|---|---|---|---|
| 1 | Equation for $\Delta S$ | | | | $\Sigma S$ products − $\Sigma S$ reactants |
| 2 | Gibbs free energy equation | | | | $\Delta G = \Delta H - T\Delta S$ |
| 3 | Condition for a reaction to be feasible | | | | $\Delta G \leq 0$ |
| 4 | Units of $\Delta G$ | | | | kJ $mol^{-1}$ |
| 5 | Convert $\Delta S$ to use with $\Delta H$ in kJ | | | | ÷ 1000 |
| 6 | Temperature at which a reaction becomes feasible | | | | $T = \Delta H \div \Delta S$ |
| 7 | Gradient of a $\Delta G$ against $T$ graph | | | | $-\Delta S$ |
| 8 | y-intercept of a $\Delta G$ against $T$ graph | | | | $\Delta H$ |
| 9 | Sign of $\Delta S$ when moles of gas increase | | | | Positive |
| 10 | Why a feasible reaction may not happen | | | | High activation energy |

*Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.*